

Can we stably earn NMR from TC?

About what I did in Numerai Tournament last year.

Ageonsen, 2023/04/08

My conclusion

Yes and No.

- High TC performance needs low correlation with Meta model.
- Strategies in Numerai tournament in my thought is the followings:
 - Create model which improve Numerai fund its own.
 - Focus on correlation.
- We might need another features to lower the volatility of TC.

What is True Contribution(TC)?

The essential metric to improve Numerai fund!

- True Contribution is the metric of how much the prediction or model contribute to Numerai fund.
- The basic idea is leave-one-out, that is how much the fund performance changes when the prediction does not exist.
- Why does the correlation score still exist?
 - > Not to make Numerai tournament more complicated.

TC sharpe ratio

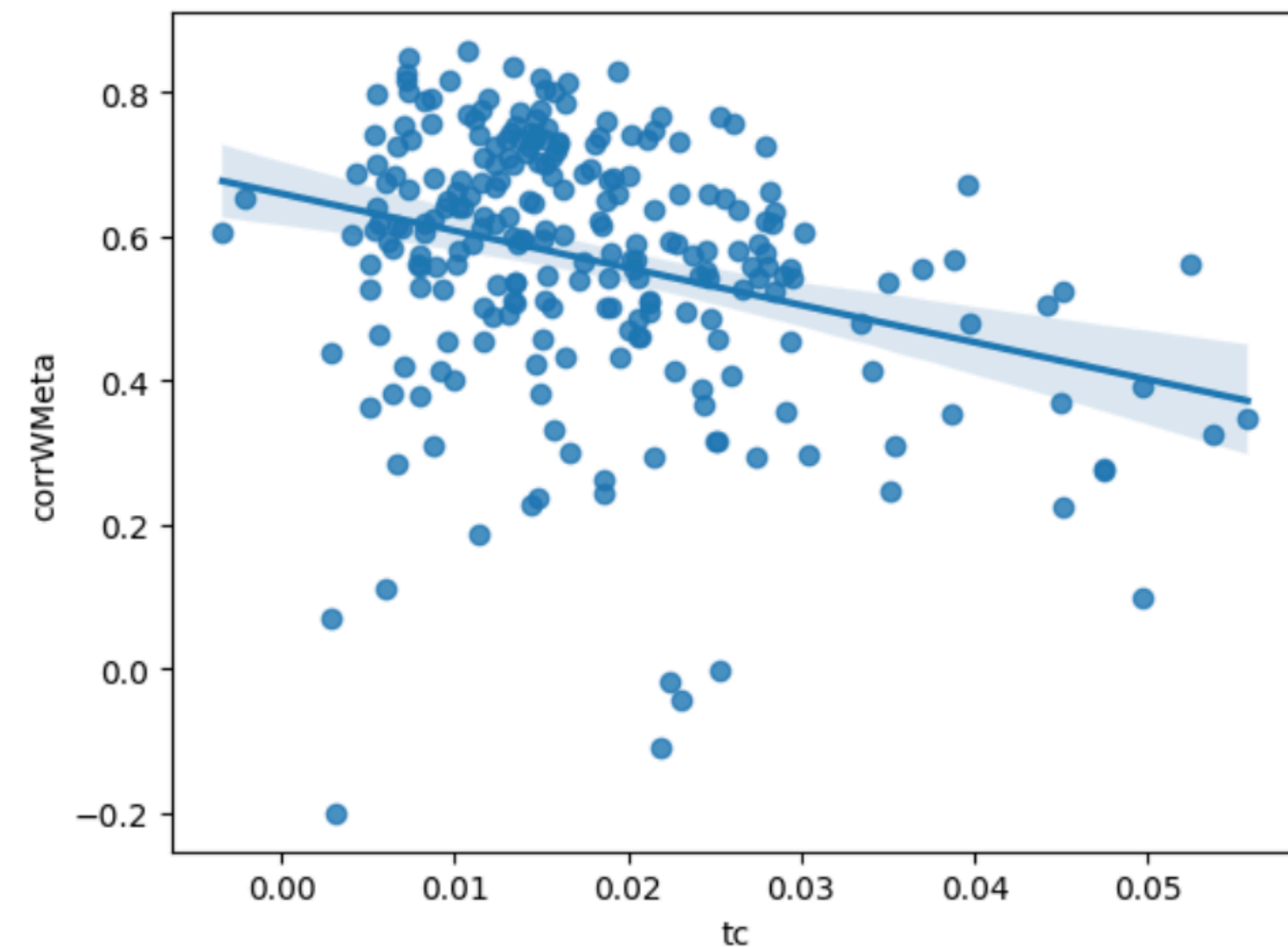
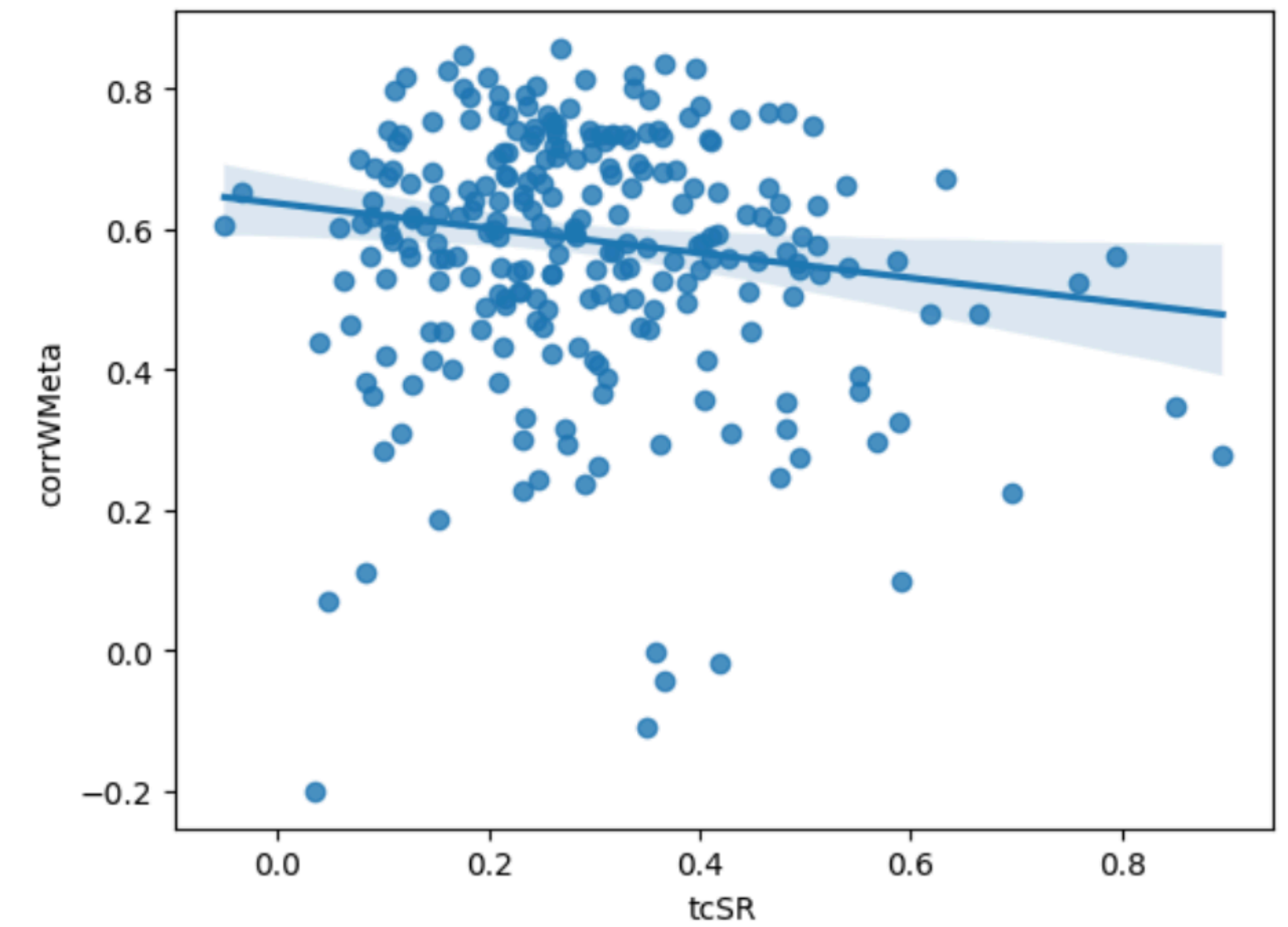
a stable TC earning metric

- I introduce the notion of TC sharpe ratio(TC_SR) as is corr sharpe.
$$\text{corr_SR} = \text{mean}(\text{corr}) / \text{std}(\text{corr})$$
- The mathematical definition of TC_SR is the following:
$$\text{TC_SR} = \text{mean}(\text{TC}) / \text{std}(\text{TC})$$
- Unlike Sharpe ratio, these definitions have no theoretical basis like CAPM that plays the central role in the modern portfolio theory.
(“Modern” portfolio theory is really modern??)

TC vs correlation with Meta model

Slightly negative regression coefficient.

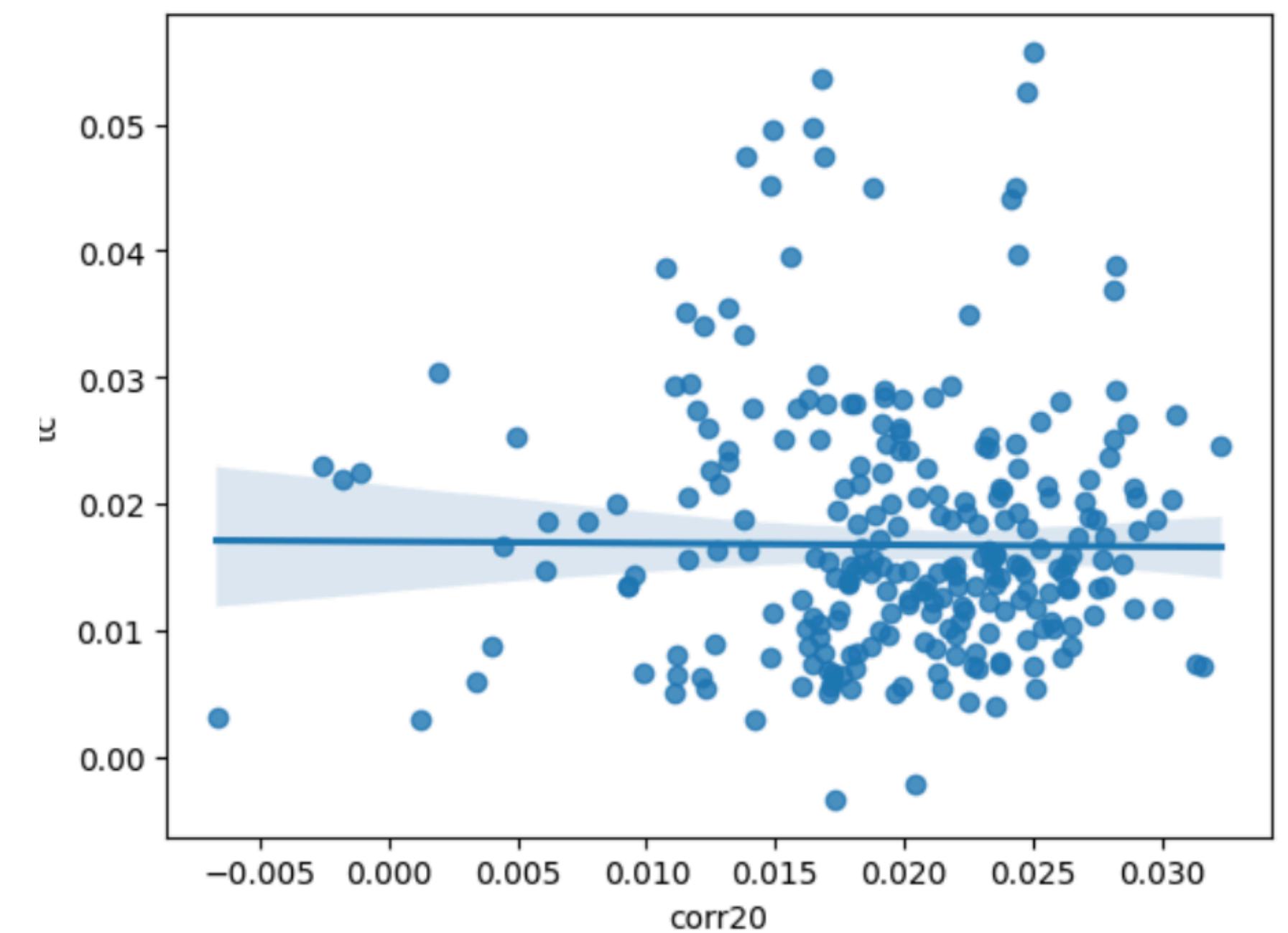
- Collect the historical performances of top 250 users such as corr20, tc, and corr with Meta model (corrWMeta)
- The lower correlation with Meta model is, the higher TC is. So is TC_SR.
- TC plays a good role for Meta model improvement.



TC vs corr20

Seems no relation between them. TC is different metric from correlation with target(target_nomi 20 days).

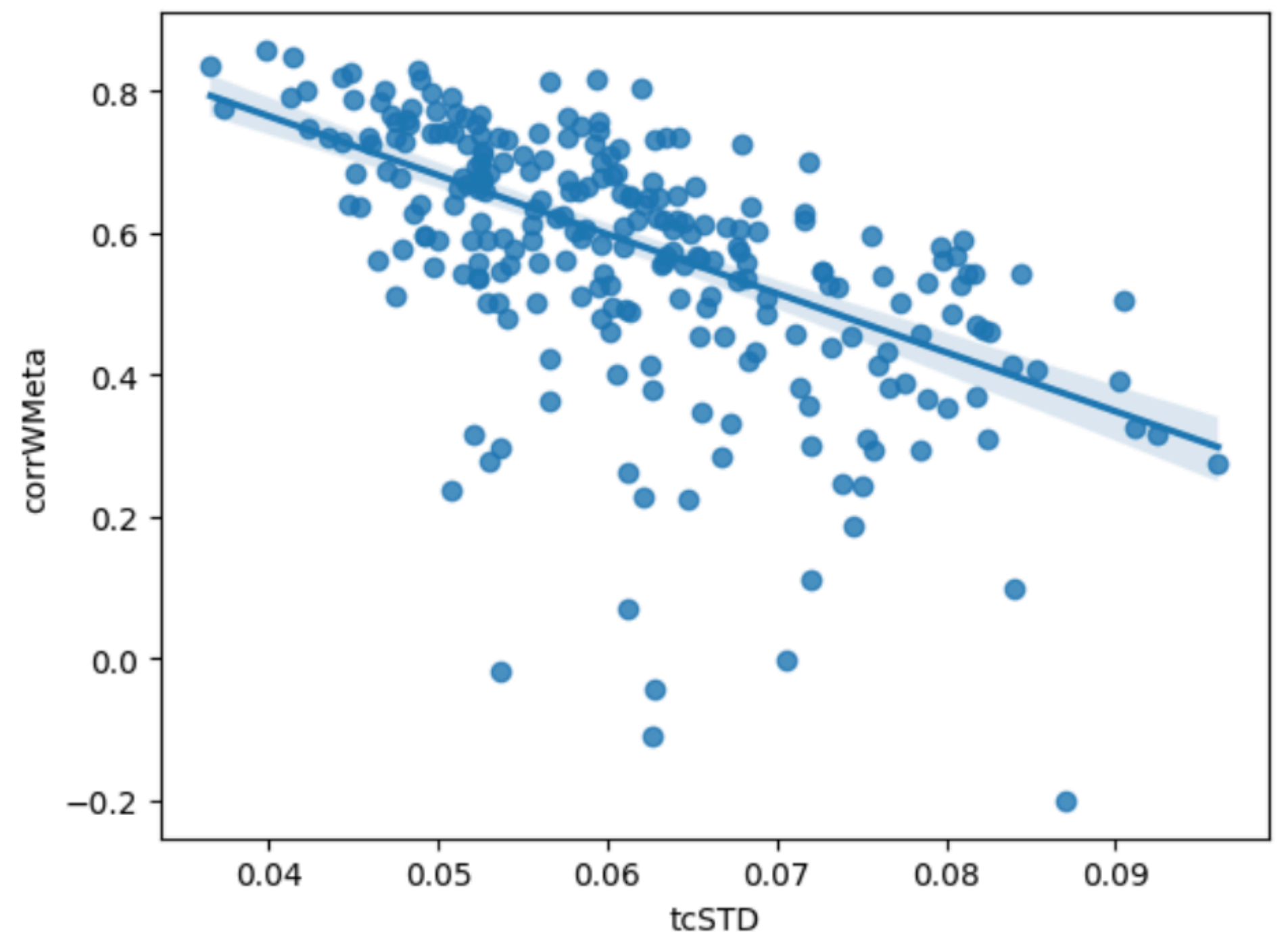
- Nearly zero correlation between TC and correlation with target, corr20.
- TC is an essentially different metric from corr20 and incentives model uniqueness for Numerai fund improvement.
- corr20 is the metric to avoid significant changes of Meta model.



TC std vs correlation with Meta model

High correlation with Meta model leads to low volatility of TC

- TC seems stable if the prediction has high correlation with Meta model.
- I want TC to have low volatility **for my mental stability**.
- I have focused on high TC with high correlation with Meta model!



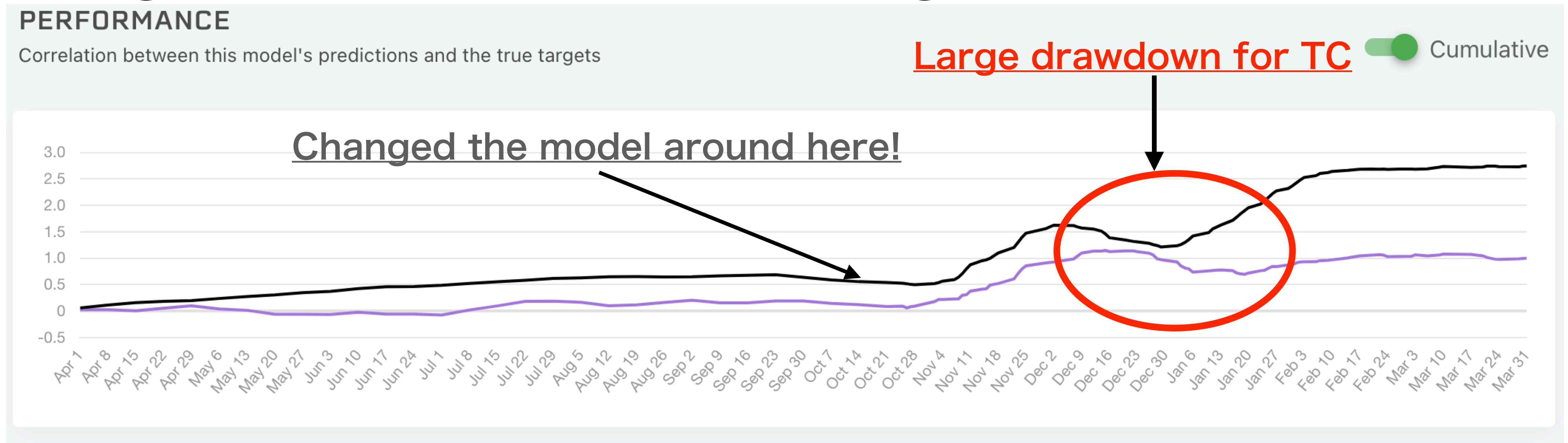
My approach

Improve corr20 by ensemble of different target model

- Models by 'target_jerome_60' certainly improve corr20 by ensemble with models by 'target_nomi_20' in my experience.
- This is mentioned in Fireside chat by Richard.
- This approach makes the predictions different from Meta model.
- By live test of different predictions created by this method, I checked the model of high correlation with Meta model.
- $\text{corrWMetamodel} \doteq 0.80$ seems appropriate.

Result

Get high corr20, but TC is not so good.

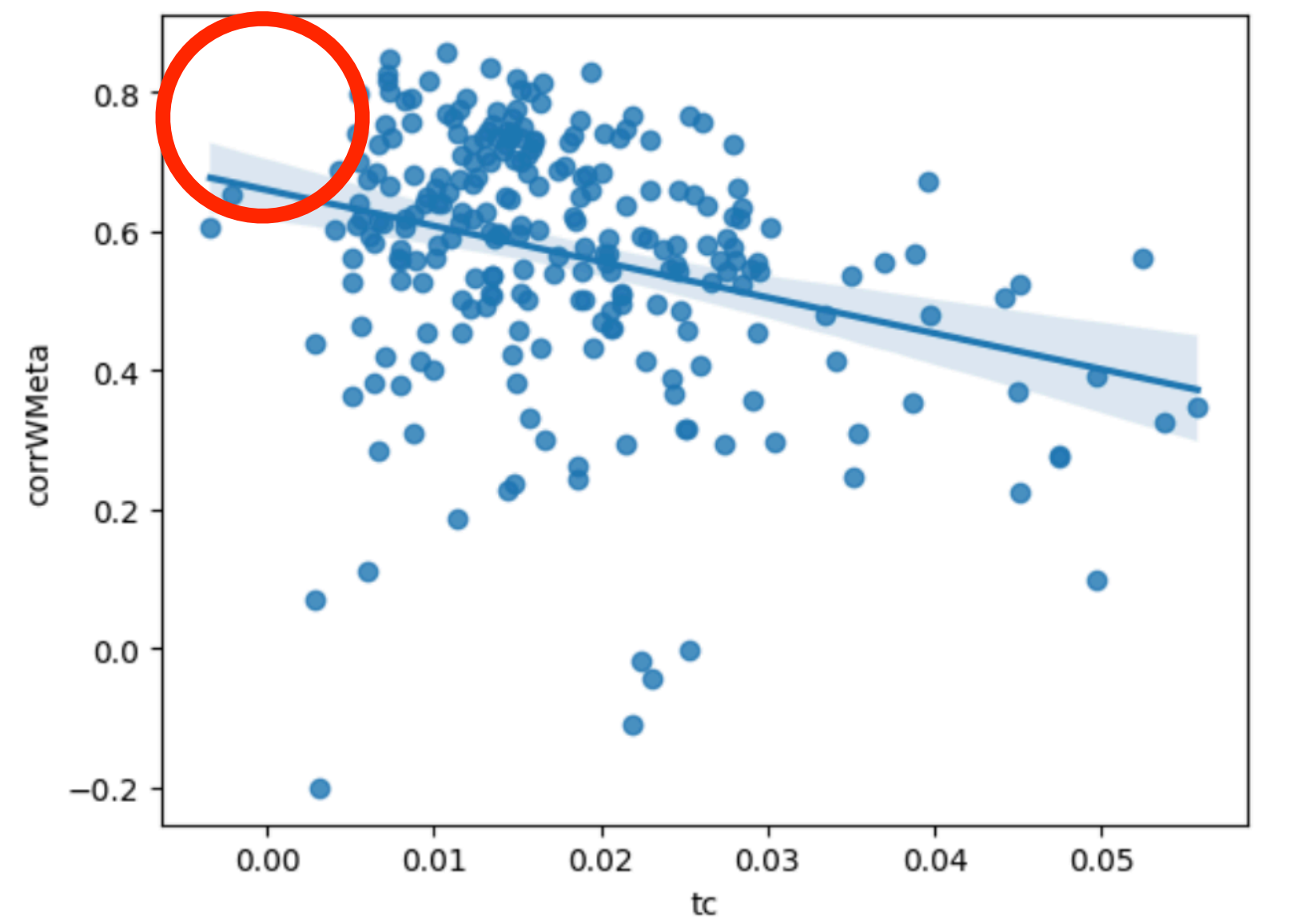
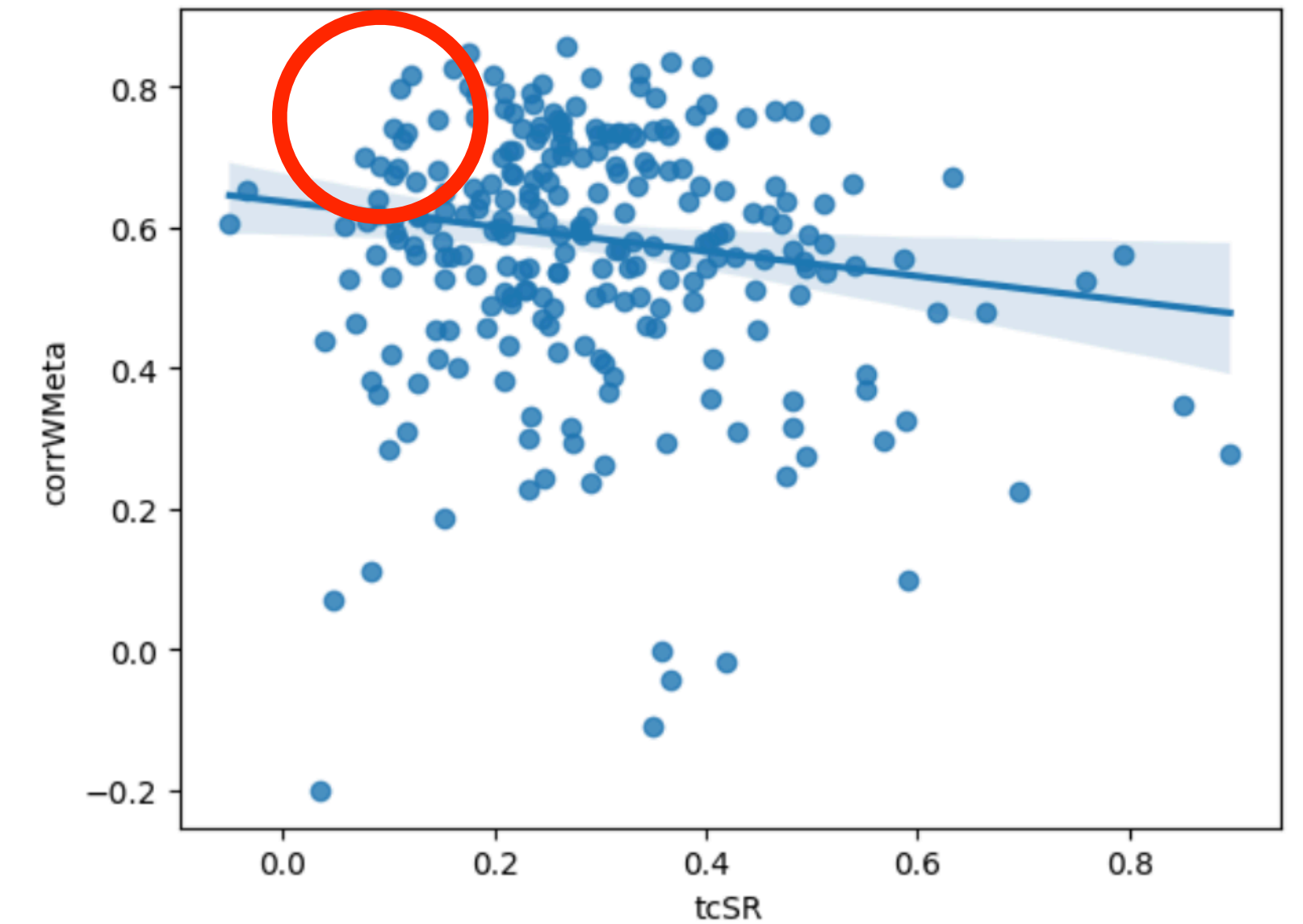


- The figure above seems great improvement of the performance.
- TC drawdown is large compared to that of corr20.

My score for the last 52 weeks

Not so good.

- My TC_SR = 0.12
- My TC = 0.0048
- corrWMetaModel = 0.79
- corr_SR = 0.60



Conclusion

TC improvements need low corrWMetamodel

- High TC/TC_SR needs low correlation with Meta model.
- There are two strategies:
 - TC improvement by ML model and target engineering.
 - improve CORR only with high corrWMetamodel
- Portfolio management of strategies by adjusting staking weight.
(Wen staking 3.0?)

Omake

My first TC staking result.

- I staked TC when the model performs well even for TC.
- The drawdown has come!!!
- TC is dangerous for me!

